Proposed Extension: Adaptive AI Response Styles

Our contribution beyond the reviewed paper (Rashid Hussain, Uzair Ahmed, Suffyan Asghar)

Motivation: The Research Gap

The reviewed paper found that **Standard**-style AI responses give the best short-term balance of low cognitive load and high task accuracy. However, it only measured a *single exposure* and did not test what happens with **repeated, long-term use**. Consistently low-effort, high-accuracy help may encourage **over-reliance** and **weaken students' independent reasoning** over time. This is the gap our own study addresses.

Our Central Question

Can we preserve the short-term benefits of low-load, high-accuracy AI responses while preventing the potential long-term cost of over-reliance and reduced independent reasoning?

What We Will Actually Do

RQ1 (Does a real AI system replicate the effect?): Test whether an actual AI chatbot (not the Wizard-of-Oz responses used in the original study) preserves the same cognitive-load benefits of Standard-style responses.

Method: Compare **No-AI vs. real Standard-style AI chatbot**. *Metrics:* mental effort, task accuracy.

RQ2 (Does repeated use cause over-reliance?): Test whether continued exposure to Standard-style AI assistance across multiple sessions increases reliance on the AI and reduces students' ability to reason independently.

Method: Same students, **4–5 sessions**, tracking actual behavior over time. *Metrics:* independent attempt rate, AI acceptance rate, independent post-test accuracy.

RQ3 (Can an adaptive AI prevent this?): Test whether an AI that adapts its response style (e.g., gradually reducing guidance as a student improves) can keep cognitive load low while still preserving independent reasoning.

Method: **Standard AI vs. Adaptive AI**, randomly assigned. *Metrics:* mental effort, AI-assisted accuracy, independent attempt rate, independent post-test accuracy.

In short: RQ1 confirms the original finding holds with a real system; RQ2 checks whether the finding backfires over time; RQ3 tests our proposed fix.